

Predictive Mobile Refueling for Agricultural Machinery via Deep Reinforcement Learning

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Abstract

With the advancement of agricultural modernization, agricultural machinery is widely used for crop harvesting. Traditionally, agricultural machines must be refueled at gas stations regularly, affecting the harvesting efficiency. A mobile refueling service has emerged in recent years, in which refueling tankers can move to serve the refueling request. However, the current mobile refueling system is still in an on-demand mode, which may not achieve timely response. Therefore, in this paper, we propose a new mobile refueling mode, i.e., predictive mobile refueling. To tackle the challenge of sparse rewards in predictive mobile refueling, we develop a two-stage reinforcement learning-based scheduling strategy MobRef, which decouples the scheduling process into a central request dispatcher and a distributed tanker reposition scheduler, and further introduces a potential energy-based reward shaping function to facilitate the training of the reposition scheduler. Extensive experiments on two real-world datasets demonstrate the effectiveness of MobRef, which outperforms the best baseline by 12.71% on average. We also present a deployed system based on MobRef, which is used internally in China National Petroleum Corporation.

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CCS Concepts

• Information systems → Spatial-temporal systems.

Keywords

mobile refueling, reinforcement learning, agricultural machinery

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1 Introduction

With the rapid advancement of agricultural modernization, the wide adoption of agricultural machinery has significantly improved the efficiency of crop harvesting. However, due to the limited fuel tank capacity of agricultural machinery, it has to interrupt the operation and go to the gas station to refuel from time to time, which would lead to reduced operation efficiency.

Realizing such inconvenience, a new refueling service, i.e., mobile refueling, emerges in recent years, e.g., Jacobus Energy¹ and Diesel Direct². In mobile refueling, when the fuel volume of agricultural machinery is running out, its operator can send a refueling request, and then the refueling tanker will come to the location of the agricultural machinery to complete fuel replenishment. However, such *on-demand* mode still suffers from the inefficiency issue.

¹<https://jacobusenergy.com/>

²<https://dieseldirect.com/>

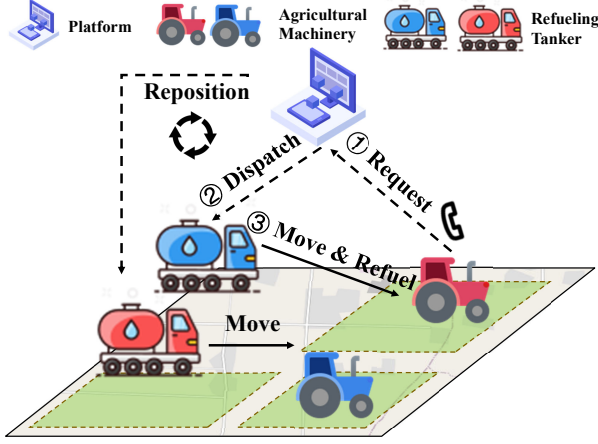


Figure 1: Predictive Mobile Refueling.

For example, if the fuel runs out before the refueling tanker arrives, the operation efficiency of agricultural machinery would be affected.

Intuitively, if the refueling tanker can move to nearby locations before the request is actually sent, the timeliness of response can be improved. Therefore, in this paper, we introduce a novel mobile refueling mode, i.e., *predictive mobile refueling*, to further boost the refueling efficiency, which is illustrated in Figure 1. It is similar to on-demand mobile refueling except that it would also periodically reposition unassigned refueling tankers to different locations expecting to serve future requests timely.

However, effective predictive mobile refueling is non-trivial due to the high dynamics of refueling requests and large action space. Ride-hailing, which is a widely studied problem, share some similarities with predictive mobile refueling if the platform can control the movement of idle drivers. In such setting, order dispatching and reposition are jointly optimized via multi-agent reinforcement learning (MARL) [25, 26]. However, we find the requests generated by agricultural machinery are sparse both in space and time, under which case MARL is difficult to learn an effective scheduling strategy due to the sparse reward signals.

Fortunately, different from ride-hailing, the mobile refueling platform knows the real-time locations of all agricultural machines (including those have not sent refueling requests), which brings us vital clues to reposition refueling tankers even when requests are not observed. Inspired by this, in this paper, we propose a *Mobile Refueling* framework, i.e., MobRef, which contains a centralized request dispatcher and a multi-agent-based tanker reposition scheduler, to schedule refueling tankers. In MobRef, the training of reposition scheduler are decoupled from the request dispatcher, which no longer relies on the sparse rewards from the requests but the spatial distributions of agricultural machines to tackle the issue of sparse rewards.

In addition, we introduce a potential energy-based reward shaping function for the reposition scheduler to further mitigate such issue.

- We propose a novel refueling mode for agricultural machinery, i.e., predictive mobile refueling.

- We devise a deep reinforcement learning-based scheduling strategy MobRef, which decouples the training of order dispatcher and reposition scheduler and is equipped with a potential energy-based reward shaping, to tackle the sparse rewards.
- Extensive experiments on two real-world datasets demonstrate the effectiveness of MobRef, which outperforms the best baseline by 12.71% on average.
- We present a deployed system based on MobRef, which is used internally in China National Petroleum Corporation (CNPC)³.

2 Preliminary

In this section, we introduce the basic concepts and formulate the predictive mobile refueling problem.

2.1 Definitions

Definition 1 (Study Region). A study region is a rectangular geographical region, where agricultural machines and refueling tankers are working within the area. It is divided into uniform grids.

Definition 2 (Farmland Region). A farmland region is a patch of agricultural fields. A study region contains several farmland regions, which indicates some grids are marked as farmlands.

Definition 3 (Refueling Tanker). The refueling tanker carries the fuel for agricultural machines and can refuel them when the tanker moves to their locations. Since roads inside farmland regions are usually field paths and roads outside farmland regions are better constructed, e.g., primary or secondary roads, we set the moving speed of refueling tanker inside / outside farmland regions as v_{in} and v_{out} , respectively. At the beginning of the day, all refueling tankers are located at a fixed location, e.g., a gas station. Each refueling tanker has a unique ID, and the platform knows its real-time location loc_t^{RT} . We denote the total number of refueling tankers in the scheduling system as N^{RT} .

Definition 4 (Agricultural Machine). Agricultural machines work within farmland regions and may also move from one farmland region to another, e.g., gathering crops. The operator of an agricultural machine may send a refueling request to the platform when he/she finds the remaining fuel is limited. When the request is sent, the agricultural machine can still move according to the working plan until the fuel runs out, which means the generated requests are also mobile, as they move along with the machine. Each agricultural machine has a unique ID, and the platform knows its real-time location loc_t^{AM} . We denote the total number of online agricultural machines as N_t^{AM} at time step t .

2.2 Problem Formulation

To find an effective refueling tanker scheduling strategy, we have an optimization time horizon, i.e., time period. It is divided into discretized time steps, and at each time step, the platform can instruct the idle refueling tankers to serve unhandled requests or move to some other grids.

Problem Statement. Given a set of refueling tankers in a study region and real-time locations of agricultural machines, predictive mobile refueling aims to find a tanker scheduling strategy at each

³<https://www.cnpc.com.cn/en/>

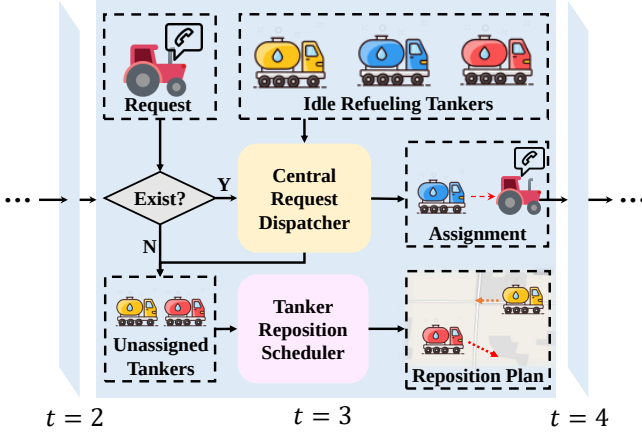


Figure 2: Overview of MobRef.

time step that can maximize the following Regularized Profit $\mathcal{J}$ over the time period:

$$\mathcal{J} = w_S \cdot S - w_C \cdot C - w_{\text{waiting}} \cdot W \quad (1)$$

where S is the total fuel supply by all refueling tankers during the period. C is the total moving cost of refueling tankers during the period. W is the summation of waiting time of all agricultural machines due to fuel running out. w_S , w_C and w_{waiting} are constant weights.

3 Methodology

Since the requests generated by agricultural machinery are highly dynamic and their movements are also stochastic, in this section, we propose a reinforcement learning-based tanker scheduling strategy, i.e., MobRef, to maximize the objective, which is adept at making decisions in the stochastic environment to maximize the long-term reward.

3.1 Overview

Our proposed MobRef adopts a two-stage framework to decouple two types of decisions, i.e., at each time step, we first use a centralized agent to make the request dispatching decision, and then reposition remaining idle refueling tankers using decentralized agents trained with reward shaping. More specifically, the architecture of MobRef is shown in Figure 2, where the decision is periodically made. At each time step, we first check whether there exist unhandled requests. If exist, we call *Central Request Dispatcher* to assign the request to an idle refueling tanker at the current time step, then for other unassigned refueling tankers, we call *Tanker Reposition Scheduler* to guide their movements. If there is no request, we directly call *Tanker Reposition Scheduler* for all idle refueling tankers. In the following subsections, we detail those two modules and elaborate the training process.

3.2 Central Request Dispatcher

The central request dispatcher takes the unhandled refueling request at the current time step, and assigns an idle refueling tanker

to serve it. A common approach is to construct a tanker-request bipartite graph for the current time step, the edge weight of which can be derived from the individual profit or estimated by a neural network [23, 26, 35], and then solve the optimal matching via the Kuhn-Munkres (KM) algorithm [11]. However, based on our analytics of the real-world data, the requests are rather limited, which is nearly impossible that concurrent requests are sent at a time step (a time step is set to a small value to achieve timely response). Therefore, the central request dispatcher is only required to simply select a tanker from all idle refueling tankers to serve the request. To this end, we can directly solve the request dispatching by a neural network as shown in Figure 3(a), which can be trained via reinforcement learning. The Markov Decision Process (MDP) is defined as follows.

State. The state s_t^d of the central dispatcher consists of the refueling tanker information RT_t , the agricultural machinery information AM_t , the mutual information MI_t of possible matched pairs and time t , i.e., $s_t^d = (RT_t, AM_t, MI_t, t)$. RT_t represents the status of all tankers, including their current locations, assignment status and intended destinations. If a tanker has been assigned to an agricultural machine that has sent a request, its assignment status is 1 and intended destination is set to the agricultural machine's current location; otherwise, its assignment status is 0 and intended destination is the same as its current location (indicating that it has not yet been assigned). AM_t represents the status of all agricultural machines, composed of locations, historical records, and request status. The historical records include the accumulated working time of the agricultural machinery, the time of the last refueling, and the refueling amount. The request status includes the requesting time (set to -1 if an agricultural machine doesn't send a request) and whether the request is responded. MI_t represents the mutual information between refueling tankers and agricultural machines, denoting the total path length and the extra length required when a refueling tanker is assigned to serve an agricultural machine. t is the time feature, represented by the time step, indicating the current moment of the day.

Action. The assignment from a request to an idle refueling tanker is regarded as an action. Since long-term reward maximization might also be achieved if we do not serve some intermediate requests, we also allow "no assignment" action, i.e., not serving unhandled requests at the current time step. To naturally incorporate the no assignment action into our model, we first assume that all refueling tankers are available for selection. When the policy network selects a matched pair where the refueling tanker is busy, we regard this scenario as a no-assignment. After an action is taken, the tanker will move to the location of the agricultural machine sending the request based on the shortest path. Note that, the agricultural machine may still continuously move in space. Therefore, at each time, we need to recalculate the shortest path to adjust the movement of the tanker to serve the request.

Reward. The immediate reward r_t^d is defined as $r_t^d = w_S \cdot S - w_C \cdot C - w_{\text{waiting}} \cdot W_1 - w_{\text{waiting}} \cdot W_2$, where first three terms represent the regularization profit obtained from the corresponding agricultural machine of this dispatch action, while the last term W_2 denotes the waiting penalty for the remaining unhandled agricultural machines

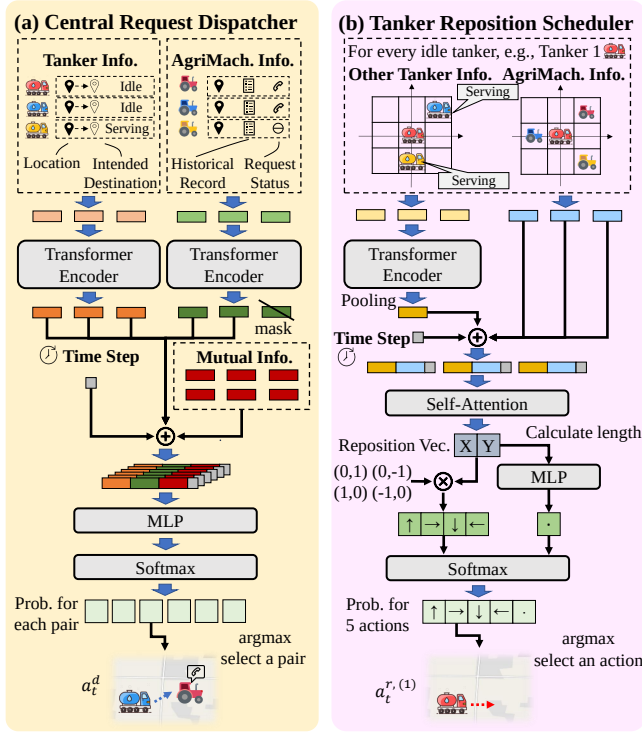


Figure 3: Detailed Implementations.

in the environment when this action is the current final action. Since the profit—which is affected by the refueling volume, refueling route, and waiting time of agricultural machinery—cannot be immediately determined at the time of assignment, we adopt a delayed reward mechanism, reserve slots for immediate rewards in the trajectories generated by the MDP, and write back the corresponding profit after the completion of dispatch actions. We can do this because our training begins only after an entire episode has finished. As for the waiting penalty of the remaining unhandled agricultural machines, the penalty at the current time step is calculated at each time step and added to the reward corresponding to the last executed action.

We use a deep neural network f_θ^d as the policy network, as shown in Figure 3(a), to learn the action probabilities of the formulated MDP, which takes the current state s_t^d as input, and gives the probabilities of each possible action $\pi_\theta^d(a | s_t^d) = f_\theta^d(s_t^d)[a]$ where $[a]$ denotes the corresponding component of the output probability vector. Since the number of active agricultural machines and refueling tankers may vary across time, to encode moving objects with varying lengths and capture their dependencies, we leverage two Transformer [29] encoders (without position embedding) to encode the states of tankers and agricultural machines, respectively, which is proven to be effective for modeling dependencies among a set of objects [21, 32, 33].

After generating the hidden states of N^{RT} tankers and N_t^{AM} agricultural machines, we only keep the states of agricultural machines with unhandled requests via masks. Then, we perform Cartesian product for the remaining states in two sets, and treat the concatenation of each pair of hidden states as well as the mutual information

MI_t and the current time step t as the hidden representation of an assignment. After that, those representations of candidate assignments would be fed into an MLP with a softmax layer to obtain action probabilities. Finally, the assignment with the maximum probability is selected as the most appropriate assignment.

3.3 Tanker Reposition Scheduler

The tanker reposition scheduler guides the movement for idle refueling tankers at each time step, such that a sent request can be timely responded in the future. In order to effectively learn the movement and request patterns of agricultural machinery, we develop another reinforcement learning-based method to reposition idle tankers. Given that the reposition of tankers is arbitrary over the whole study region, a centralized agent for reposition is infeasible. Therefore, we develop a multi-agent-based model for tanker reposition as shown in Figure 3(b).

3.3.1 Basic Implementation. We regard the tanker reposition as another Markov Decision Process (MDP) and treat each tanker as an agent, which is formulated as follows.

State. The state of each agent $s_t^{r,(i)}$ consists of other tanker information OT_t , the agricultural machinery information AM_t' , and the time step t , i.e., $s_t^{r,(i)} = (OT_t, AM_t', t)$, which is observed from the perspective of the decision-making refueling tanker agent. OT_t and AM_t' are nearly identical to RT_t and AM_t used in the central request dispatcher, except that OT_t excludes refueling tanker info of the agent itself and absolute locations are also replaced with relative locations as the agent's observation.

Action. For each agent, we define 5 reposition actions, i.e., moving up, down, left, right, and standing still. After an action is selected, the tanker will move toward the direction in the current time step.

Reward. Since we decouple the request dispatching and tanker reposition, in order to make the network trainable, we assume if a tanker moves to the location of an agricultural machine with unhandled request, it can refuel the machine. Based on such assumption, the profit of a tanker is $p_t^{env,(i)} = w_S \cdot S_t^{(i)} - w_C \cdot C_t^{(i)} - w_{waiting} \cdot W_t^{(i)}$, which represents the refueling reward, fuel consumption penalty, and agricultural machinery waiting penalty in the environment (allocated equally to each agent, respectively). Each agent has its own MDP trajectory, with the goal of maximizing its respective cumulative reward; thus, we have $\hat{r}_t^{r,(i)} = p_t^{env,(i)}$.

Another deep neural network f_ϕ^r , as shown in Figure 3(b), is used to learn the policy of the formulated MDP, which takes the current state $s_t^{r,(i)}$ of a tanker as input, and gives the probabilities of each possible direction $\pi_\phi^r(a | s_t^{r,(i)}) = f_\phi^r(s_t^{r,(i)})[a]$. Essentially, this function weights the relative position vectors of agricultural machines to enable repositioning. OT_t is encoded by a Transformer Encoder and pooled into a context vector, which is then concatenated with AM_t' and t . Self-attention scores computed from the concatenated features are used to weight the relative position vectors of individual agricultural machines with respect to the current repositioning agent, yielding the reposition vector. This vector is dotted with four directional unit vectors to generate movement

scores, while its length is fed into an MLP to produce the stay score. Finally, softmax is applied to all five action scores to obtain the action probability distribution.

3.3.2 Potential-based Reward Shaping. Though the score network can be learned based on the aforementioned reward function in theory, unlike request dispatching, where each assigned tanker has a clear refueling target, refueling purely based on reposition can hardly receive positive rewards due to the large action space and limited requests. Therefore, in order to facilitate the training of the reposition scheduler, we modify the reward function by introducing the reward shaping technique [19].

Main Idea. Reward shaping aims to introduce additional rewards to accelerate the learning of agents. The principle is to make positive rewards denser to mitigate the reward sparsity issue. In our scenario, intuitively, if an idle refueling tanker is closer to an agricultural machine, it may response to its potential request timely in the future.

Implementation. To achieve the above idea, we first introduce a spatial proximity function D between a refueling tanker i and an agricultural machine j :

$$D(d_{ij}) = e^{-w_d \cdot d_{ij}} \quad (2)$$

where d_{ij} is the geographical distance between a tanker i and a agricultural machine j , and w_d is a hyperparameter to adjust the smoothness.

Furthermore, we treat the spatial proximity as the potential energy E_{ij} of the agricultural machine to the refueling tanker:

$$E_{ij} = D(d_{ij}) \quad (3)$$

Now, we can calculate the potential energy $E(s_t^{r,(i)})$ of tanker i by summation over impacts of online agricultural machines:

$$E(s_t^{r,(i)}) = \sum_{j=1}^{N_t^{AM'}} E_{ij} \quad (4)$$

where $N_t^{AM'}$ denotes all online agricultural machines that have not submitted any requests. We exclude the agricultural machines that have sent requests based on the following considerations: assuming that our central request dispatcher has learned correctly, when there exist unhandled requests and idle tankers in the reposition stage, the dispatcher must take a no-assignment action, expecting to allocate the request to tankers that will become idle in the future (i.e., those currently in service). As for idle tankers at present to be repositioned, the requesting agricultural machines should not contribute to the potential energy.

We define the potential-based reward shaping function as the potential energy difference between the two consecutive time steps:

$$r_t^{pe,(i)} = E(s_{t+1}^{r,(i)}) - E(s_t^{r,(i)}) \quad (5)$$

Finally, we place the original immediate reward function with the combination of the actual rewards from the environment as well as the potential-based rewards adjusted by the weight hyperparameter w_r , as follows:

$$r_t^{r,(i)} = p_t^{env,(i)} + w_r \cdot r_t^{pe,(i)} \quad (6)$$

where w_r serves to balance the relative ratio between the two reward terms, owing to their different magnitude scales. To achieve

the desired effect, these two scales need to be carefully coordinated: this way, the reposition action of refueling tankers can be correctly guided by our designed potential function, while also enhancing environmental rewards.

3.4 Training Process

We adopt a phased training process to ensure the learning stability as many stage-based RL frameworks did [35] to prevent the central request dispatcher and the tanker reposition scheduler from interfering with each other. More specifically, we first train the central request dispatcher, and then train the tanker reposition scheduler, to let the tanker reposition module adapt to the dispatch strategy. We use REINFORCE [34] with a policy network to train both the central request dispatcher and tanker reposition scheduler. The pseudocode of our training process is given in Algorithm 1.

To train central request dispatcher, for each time step, if there exists unhandled requests, we employ the dispatch strategy π_θ^d based on the action probability to select an assignment a_t^d to execute. Once the requested agricultural machine is refueled, we obtain the regularized profit for executing the action, and add it back to the reward corresponding to the action in the MDP trajectory. Finally, leveraging the Monte Carlo episodic trajectories sampled under the dispatch strategy π_θ^d , θ can be optimized via gradient ascent based on the following policy gradient formula:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta^d} \left[\sum_{t=0}^T \left(\sum_{t'=t}^T \gamma^{t'-t} r_{t'}^d \right) \nabla_\theta \log \pi_\theta^d(a_t^d | s_t^d) \right] \quad (7)$$

where $\pi_\theta^d(a_t^d | s_t^d)$ essentially is the output of $f_\theta^d(s_t^d)$ when the action is a_t^d , representing the probability. T is the total length of episodic trajectory. $\sum_{t'=t}^T \gamma^{t'-t} r_{t'}^d$ is the accumulated discounted reward from time step t .

After central request dispatcher is trained, we now learn tanker reposition scheduler π_ϕ^r . At this phase, parameters in central request dispatcher are frozen. At each time step, we first handle requests using the strategy π_θ^d , selecting the action with the maximum probability. After that, if there exist idle tankers, we employ the strategy based on the action probability to select a reposition action $a_t^{r,(i)}$ for each agent i , i.e., idle refueling tanker, to execute. Our agents adopt the method of sequential decision-making, that is, after one agent makes a decision, other agents update their observations. Once the action is completed, we obtain the final state and summed rewards for executing the action. Finally, leveraging the Monte Carlo episodic trajectories sampled under π_ϕ^r , ϕ can be optimized via gradient ascent based on the same policy gradient formula.

4 Simulator

Since RL operates through trial-and-error, a dynamic simulation environment is essential for proper training and evaluation. Therefore, we construct a simulator based on the real-world data.

We obtain the real-world trajectories of agricultural machinery from the National Agricultural Machinery Operation Command and Dispatch System⁴. Based on those data, the simulator can simulate

⁴<https://e.camtc.cn/>

Algorithm 1: Training Process

Input : Training episodes for request dispatcher and reposition scheduler N^d, N^r ; time period T ; environment env .

Output: Trained policy network π_θ^d and π_ϕ^r .

```

/* Central request dispatcher training. */
1 for episode  $\leftarrow 1$  to  $N^d$  do
2   Empty MDP trajectory  $\tau^d$ ;
3   for  $t \leftarrow 1$  to  $T$  do
4     if  $env.request() \neq \emptyset$  then
5        $a_t^d \leftarrow \text{Sample from } \pi_\theta^d(a | s_t^d)$ ;
6        $env.assign(a_t^d)$ ;
7       Initialize  $r_t^d$  as empty;
8       Store  $s_t^d, r_t^d, a_t^d$  into the trajectory  $\tau^d$ ;
9       for tanker  $i$  in all tankers do
10        if  $env.status(i)$  is "complete the assignment this step" then
11          Get the time step  $t'$  when the assignment is made, and calculate the regularized profit  $\mathcal{J}_{t'}$  of the action.
12           $r_{t'}^d \leftarrow r_{t'}^d + \mathcal{J}_{t'}$ 
13      Calculate the waiting penalty  $\mathcal{P}_t$  within the time step for agricultural machines with unhandled requests;
14       $r_t^d \leftarrow r_t^d + \mathcal{P}_t$ 
15      Calculate policy gradient  $\nabla_\theta J(\theta)$  based on  $\tau^d$ ;
16      Update  $\theta$  via gradient ascent;
/* Tanker reposition scheduler training. */
17 for episode  $\leftarrow 1$  to  $N^r$  do
18   Empty MDP trajectory  $\tau^r$  of all agents;
19   for  $t \leftarrow 1$  to  $T$  do
20     if  $env.request() \neq \emptyset$  then
21        $a_t^d \leftarrow \text{argmax}_a \pi_\theta^d(a | s_t^d)$ ;
22        $env.assign(a_t^d)$ ;
23     for tanker  $i$  in all idle tankers do
24        $a_t^{r,(i)} \leftarrow \text{Sample from } \pi_\phi^r(a | s_t^{r,(i)})$ ;
25        $s_t^r, r_t^{r,(i)} \leftarrow env.repo(s_t^{r,(i)}, a_t^{r,(i)})$ ;
26       Store  $s_t^{r,(i)}, r_t^{r,(i)}, a_t^{r,(i)}$  into the trajectory  $\tau^{r,(i)}$ ;
27   Calculate policy gradient  $\nabla_\phi J(\phi)$  based on  $\tau^r$ ;
28   Update  $\phi$  via gradient ascent;
29 return  $\pi_\theta^d$  and  $\pi_\phi^r$ 

```

the behaviors of agricultural machinery, i.e., moving in farmland regions, sending refueling requests, and being refueled.

More specifically, in order to adapt to the spatio-temporal discretized setting, we first transform trajectories into discretized representation, i.e., the longitude and latitude are mapped to specific grids of the study region and timestamps are mapped to time steps. Then, we regard the trajectory of each agricultural machine within

Table 1: Statistics of Datasets.

Dataset	WheatAgri	CornAgri
Time Span	2021/6 - 2022/10	2019/8 - 2019/10
#. of Total Machines	25	73
#. of Online Machines / Hour	12	15
#. of Trajectories	425	625

each episode as its “working plan”. At the beginning of an episode, we assume all agricultural machines are fully fueled, and the fuel consumption rate of the agricultural machine is η .

We assume that the timing of agricultural machines sending refueling requests is around a remaining fuel proportion. According to practical scenarios, different drivers have different refueling preferences: some conservative drivers send refueling requests with ample remaining fuel, while others only do so when their fuel level is low. Therefore, we sample a threshold from a distribution for each agricultural machine when an episode starts, and keep the thresholds unchanged within the episode. We set the distribution as a Gaussian distribution. The mean value is set to 0.5, and the standard deviation is set to 0.1. We also limit the possible values can be sampled between 0.4 and 0.6 to avoid some extreme cases. Once the proportion of remaining fuel is lowered than the threshold, a refueling request is sent.

After the request is sent, if there is still remaining fuel, the machine can follow its working plan to move. Otherwise, it would stay in place to wait for the refueling tanker. When a tanker refuels the machine, it can continue to move following its working plan.

5 Experiments

In this section, we evaluate the performance of the proposed method on real-world datasets.

5.1 Datasets

Two real-world agricultural machinery trajectories from National Agricultural Machinery Operation Command and Dispatch System are adopted to train the model and evaluate the performance, which are sampled every second. One dataset is from Shandong, China (WheatAgri for short). The other is a public dataset⁵ distributed in 5 provinces of China including Shanxi, Shaanxi, Henan, Shandong and Anhui. Since the released dataset from each individual province is insufficient for training and evaluation, we have shifted trajectories into one study region for reproducibility (CornAgri for short). The detailed statistics are summarized in Table 1.

We report some vital characteristics of both datasets as follows. Figure 4(a) shows the cumulative distribution of the trajectory length of agricultural machinery within a day, and we can see that the maximum traveling distance of agricultural machines within a day is around 100 km. Figure 4(b) shows the cumulative distribution of the trajectory duration of agricultural machinery within a day.

We set the evaluation period/episode to one day, and the datasets are splitted into training set and testing set with a ratio of 8:2.

⁵https://github.com/Agribigdata/public_trajectory_dataset

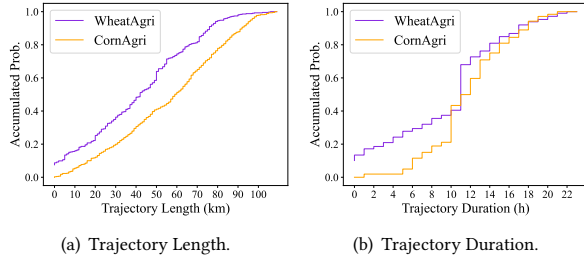


Figure 4: Data Distributions.

5.2 Experimental Settings

Baselines. Since there is no existing work to handle predictive mobile refueling, we adopt some representative methods from the most similar setting, i.e., ride-hailing, for comparison, including traditional methods and reinforcement learning-based methods.

- **Random:** After an agricultural machine sends a request, we randomly pick a refueling tanker to serve it.
- **Nearest Neighbor [8]:** After an agricultural machine sends a request, we dispatch the nearest refueling tanker to serve it.
- **KM [11]:** At each time step, we build a tanker-request bipartite graph where each edge weight is the individual predicted profit serving the request and solve the maximum matching.
- **POLAR [27]:** Using the training data, it predicts the spatio-temporal distribution of agricultural machine requests and idle refueling tankers in the test set and generates a matching plan accordingly. In practical tests, matching follows the plan, and refueling tanker repositioning is guided by the predicted agricultural machine request locations.
- **CVNet [26]:** It uses a value network to estimate the value of the idle tanker’s current state and the value of reaching the destination after responding to a request. The matching score between drivers and orders is calculated based on the difference in state values, trip revenue, and user experience, after which KM algorithm is applied to solve the matching problem.
- **JDRL [25]:** It treats every tanker as an agent, and use MARL to choose the action. The action space consists of the various directions of repositioning and the grids in the environment. Selecting a grid represents selecting a requesting agricultural machine within that grid.
- **D2SN [37]:** It is a two-layer MDP framework. The inner-layer MDP iteratively choosing to select a request-tanker pair or end assignment in the current time step. The outer-layer MDP executes all actions in the inner MDP as a joint action, and sums all corresponding immediate rewards. The learning of the policy network is guided by the probability of the joint action.

Variants. We compare MobRef with following variants:

- **MobRef w/o RS:** When training the agents of reposition, we do not use the reward shaping function proposed in Sec. 3.3, but only use the feedback of the environment itself (i.e. refueling profit) to train the agents.
- **MobRef w/o REP:** We completely remove the reposition module to see how much utility gain can be brought by the reposition module we proposed in Sec. 3.3.

Table 2: Performance under Different Numbers of Tankers.

Method	WheatAgri			CornAgri		
	$N^{RT}=2$	$N^{RT}=3$	$N^{RT}=4$	$N^{RT}=2$	$N^{RT}=3$	$N^{RT}=4$
Random	96.08	249.57	320.47	45.34	152.71	236.29
Nearest Neighbor	294.98	365.12	386.15	369.36	452.11	475.63
KM	<u>303.60</u>	365.24	387.85	373.40	448.23	480.42
POLAR	294.87	<u>370.09</u>	<u>398.84</u>	324.66	423.51	478.76
CVNet	299.31	357.87	383.79	363.77	426.69	474.50
JDRL	286.44	342.52	362.32	341.99	420.39	463.40
D2SN	301.96	365.84	385.46	<u>392.35</u>	476.05	501.41
MobRef (Ours)	343.73	413.20	437.86	464.11	543.75	547.12

- **MobRef w/o RLD:** We change the order dispatching method based on Reinforcement Learning to nearest neighbor dispatching.

Evaluation Metrics. Regularized profit $\mathcal{J}$ in Equation 1 are used for evaluation. w_s , w_C and $w_{waiting}$ are set to 0.1/L, 0.015/km, and 0.02/min, respectively according the experts’ knowledge.

Simulator Settings. Both study regions are divided into 155×185 uniform grids (grid size: 500m× 500m), and the timestep interval is set to one minute. For agricultural machinery, the default fuel tank capacity is 200L and the fuel consumption rate η is set to 12L/km.

Training Details & Hyperparameters. Our network structure mainly includes TransformerEncoder and MLP. Both the number of layers and the number of heads of the TransformerEncoder are set to 2, and the number of hidden layer units in its feed-forward network and in the MLP are both 128. The default number of refueling tankers N^{RT} is 3. The moving speed of the refueling tankers inside the farmland regions is $v_{in}=100\text{m/min}$, and outside the farmland regions is $v_{out}=500\text{m/min}$. The locations of all refueling tanker are initialized at the center of the study region at the beginning of each episode. The discount factor γ is set to 0.99.

Implementations. Our algorithms are implemented in Python with PyTorch. Experiments were conducted on a server with 56 Cores@2.60GHz, 56GB memory, and models were trained using a GeForce RTX 4090 (24GB) GPU. We have also released the code for the public use⁶.

5.3 Overall Performance

Table 2 presents the performance of all methods under different numbers of tankers. MobRef outperforms the best baselines by an average of 11.55% and 13.88% on two datasets, respectively, with an overall average performance improvement of 12.71%. Nearest Neighbor and KM achieved nearly the same performance, because the request in our scenario is relatively sparse, and there are few cases where multiple requests are generated simultaneously within a single time step. POLAR derives schedules from the training dataset, which is not fully applicable to the test dataset. Yet, its schedules exhibit good robustness when the number of tankers is large in the WheatAgri dataset, thus yielding decent performance. CVNet exhibits average performance, primarily due to the inaccuracy of its value estimation in our complex scenario. Similarly, JDRL

⁶<https://github.com/stdi-lab/MobRef>

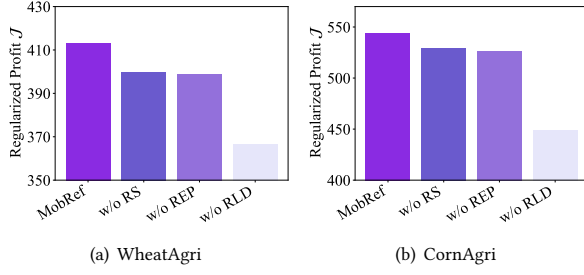


Figure 5: Ablation Study.

performs poorly, as it is challenging to jointly train the dispatch and reposition strategies. D2SN yields relatively good performance. However, its unsuitable no-assignment mechanism and the absence of a reposition strategy prevent it from attaining better performance.

We observed that the performance lead of our method narrows as the number of tankers increases, which is attributed to the shrinking optimization space under resource-sufficient conditions. We also observed from the results that the CornArgi dataset features a more dynamic and complex environment than the WheatAgri dataset, with relatively more requests sent by agricultural machinery. In such cases, the optimization space expands and the performance gaps between different methods widen. As can be seen, due to the higher request frequency and more frequent refueling events, all methods yield higher overall metric values—with the exception of Random. Its poor strategy that incurs greater waiting penalties instead. Polar performs relatively poorly on the CornArgi dataset because the more highly dynamic environment reduces the adaptability of its pre-estimated schedules. In addition, MobRef achieves a relatively larger performance lead on the CornArgi dataset.

5.4 Ablation Study

We evaluate the effect of each component of MobRef by removing each of our designs. The results are shown in Figure 5. It is worth noting that removing potential-based reward shaping and removing repositioning have similar effects because in the case of sparse rewards, using only environmental feedback rewards, the agent learns a strategy of not repositioning (i.e., tends to be stationary) for most of the time. However, it is still guided by environmental rewards, particularly in reducing the waiting penalty of agricultural machinery, and thus yields minor effects. The greatest effectiveness decrease is achieved by removing RL-based dispatching, as RL has the ability to optimize long-term returns.

We also observe that the integration of our reposition strategy into the Nearest Neighbor heuristic does not result in performance enhancement relative to the baseline without this strategy. The root cause lies in the fact that the heuristic algorithm does not implement an optimal policy corresponding to the current environmental state. Even if repositioning directs idle tankers to better locations, the heuristic-driven decision-making mechanism cannot support effective future dispatch arrangements. On the contrary, our dispatch method is capable of making good decisions across varying states and realizes notable performance improvements.

Overall, each component plays an important role in our method, and removing any one would result in a decrease in effectiveness.

5.5 Hyperparameter Study

5.5.1 Impact of w_d . Figure 6(a) shows the impact of w_d , whose main purpose is smoothing. The smaller the value of w_d , the smoother the function D . A large parameter w_d means that the closer the refueling tanker is to the agricultural machinery, the greater the immediate reward that moving one grid can bring, which means the reward function is less smooth. From the results in Figure 6(a), it can be observed that the overall performance is optimal when the value of w_d is set to $1e-2$.

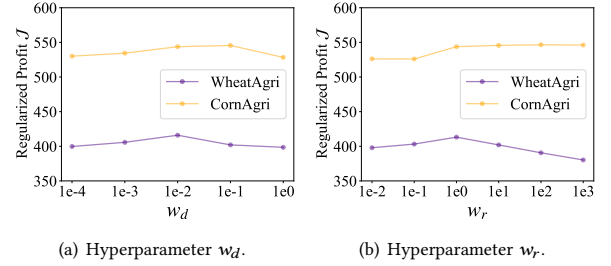


Figure 6: Hyperparameter Study.

5.5.2 Impact of w_r . Figure 6(b) shows the impact of w_r , which is the coefficient used in repositioning to balance the environmental reward and the reward-shaping function we designed. The performance of repositioning varied with different values of w_r across the two datasets. When w_r was small (e.g., 0.1 or 0.01), the reward signal was dominated by the environmental reward, rendering the potential-based reward shaping nearly ineffective. The final effect was analogous to removing the reward shaping module: most of the time, the repositioning agent kept idle tankers stationary. When $w_r = 1$, both datasets achieved favorable performance, as this value well balanced the two reward components. As w_r increased further—where the reward shaping function fully governed the reward signal—the performance of CornArgi remained largely unchanged. This indicates that the manually designed reward shaping is generally aligned with the optimization objective, which can regulate the agent’s behavior correctly even without environmental rewards. In contrast, WheatAgri exhibited a significant performance decline, suggesting that there is still a discrepancy between the guidance of our designed reward shaping and the actual optimization objective. It is thus imperative to constrain w_r within a reasonable range and regulate the overall reward signal via environmental rewards. We set $w_r = 1$ to achieve the optimal overall performance.

6 System Deployment

We deploy MobRef in China National Petroleum Corporation (CNPC), which is one of the largest oil and gas company in the world.

The system is equipped with a platform and a mobile client. The platform has access to real-time locations of agricultural machinery as well as refueling tankers in the study region using Beidou. Figure 7(a) is the web interface of the platform, which displays the number of refueling tankers and agricultural machinery operators

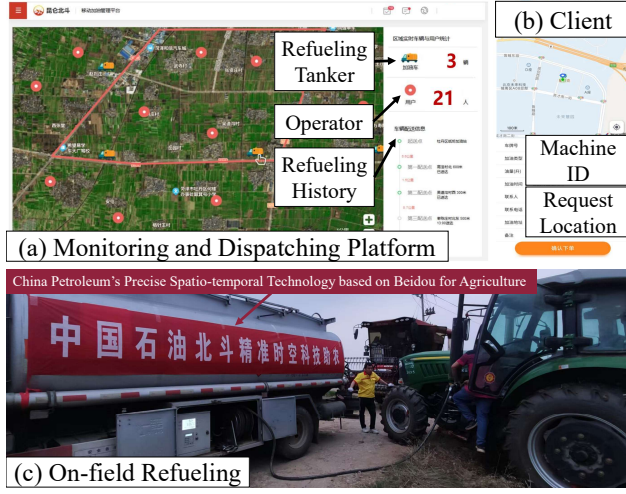


Figure 7: System Deployment.

online. When a refueling tanker is clicked, the refueling history of the tanker would be displayed. MobRef would periodically reposition the locations of idle tankers. The client is an APP installed on agricultural machinery operator’s mobile phone for sending the refueling request as shown in Figure 7(b). In the APP, the operator can fill in the agricultural machine ID and request location, then a refueling request would be sent to the platform after the orange button is clicked, and MobRef would assign an appropriate idle tanker to serve the request. When the assigned refueling tanker moves to the location of the agricultural machine sending the request, it can refuel the machine. As shown in Figure 7(c), the refueling tanker is on the left, the agricultural machine is on the right, and there is a pipe between them for fuel transferring.

We evaluate the performance of MobRef serving the refueling requests of wheat harvesters. Since the time window of harvesting wheat is quite tight, the performance of scheduling policy can be mainly evaluated in June and July every year. Therefore, we select two study regions in Shandong Province with similar scale of the offline experiments, and allocate three refueling tankers for each region. In 2023/6–2023/7, we first employ the “on-demand” policy to schedule the nearest tanker to serve the request, i.e., Nearest Neighbor baseline. And in the next year, i.e., 2024/6–2024/7, we employ MobRef to schedule refueling tankers in those two regions. A comparison of two years of data shows that profits increased by an average of around 5% after MobRef was applied. We find the improvement is smaller than the offline experiments, which might be attributed to non-perfect modeling of the simulator and variation of grain production of different years.

7 Related Work

7.1 Ride Hailing

With the popularity of shared economy, ride-hailing [6, 23, 25, 26, 36], and ride-sharing [13, 18, 30, 41] are widely studied in recent years. Ride hailing systems are typically designed to solve two problems: order dispatching and driver reposition.

In order dispatching, Li et al. [15] use the nearest neighbor methods, but a more common approach is to use bipartite graph matching [2, 6, 12, 23, 24, 36]. There are also works [9, 16, 37] that utilize RL to directly select assignments. If the platform can control the movement of idle workers, some methods like prediction-based matching [27] or RL [10, 25, 26] can be used to achieve reposition.

In our work, we perform similar order dispatching and repositioning for refueling tankers. However, we differ from ride-hailing from following three aspects: 1) the locations of potential requests are known, which is a vital clue should be considered; 2) the sent request is moving, which makes a learning-based solution essential; and 3) the moving objects are sparsely distributed, makes coupled optimization method difficult to learn a good strategy.

7.2 Decision-making Driven by ST Data

Spatio-temporal data-driven decision-making aims to adopt strategies based on spatio-temporal data [1, 8, 22], which can be categorized into static decision-making and dynamic decision-making. For example, in static decision-making, one can select locations to maximize the trajectory coverage [1], or partition the delivery zones to balance the workloads of couriers based on historical delivery data [5]. In dynamic decision-making, apart from the order dispatching, there are also various other tasks [28, 39, 40], e.g., parcel delivery [4, 7, 9, 16, 17, 38], urban crowdsensing [31], bike reposition [15], public resource allocation [20], traffic signal control [14], ambulance reposition [8], and vehicle loading plan generation [3]. In our work, we also study the problem of dynamic decision-making driven by spatio-temporal data. However, we explore a different scenario, i.e., predictive mobile fueling, which makes decisions based on historical sparse refueling request patterns and distributions of agricultural machines.

8 Conclusion

In this paper, we propose a new mobile refueling mode for agricultural machinery, i.e., predictive mobile refueling. We propose a two-stage deep reinforcement learning method MobRef to tackle it, which overcomes the sparse reward issue. MobRef is composed of a central request dispatcher and a tanker reposition scheduler. To facilitate the training of the reposition scheduler, we introduce a potential-based reward shaping function. Experiments on two real-world datasets show that our method has an average 12.71% improvement compared to the best baseline, and a system based on MobRef is deployed and used in China National Petroleum Corporation.

Although we solve the problem of training the decoupled tanker reposition scheduler under a sparse reward scenario, it may still lack a long-term perspective on the reposition due to insufficient collaboration with the dispatcher. In the future, it is worth considering designing a cooperative framework for refueling tanker reposition and request dispatching.

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